Supplementary Material

The Web Application

We provide a public web application for interactive inspection of the SoftFuse-KGC pipeline at: https://inosuke710-softfuse-kgc-demo.hf.space/. The web application is a demonstration deployed on Hugging Face Spaces and uses precomputed artifacts. It allows users to inspect validation and test examples across PrimeKG, PharmKG, Hetionet, DRKG, and repoDB. After opening the page, users select a dataset, split, disease/query case, and top-k value. The three main tabs then expose the complete demo flow: an overall prediction summary, candidate reranking by soft support, and evidence-subgraph selection by fuzzy retrieval.

Figures S1–S3 document the three tabs and identify the controls, result panels, candidate lists, graph elements, and visual encodings used in the interface.

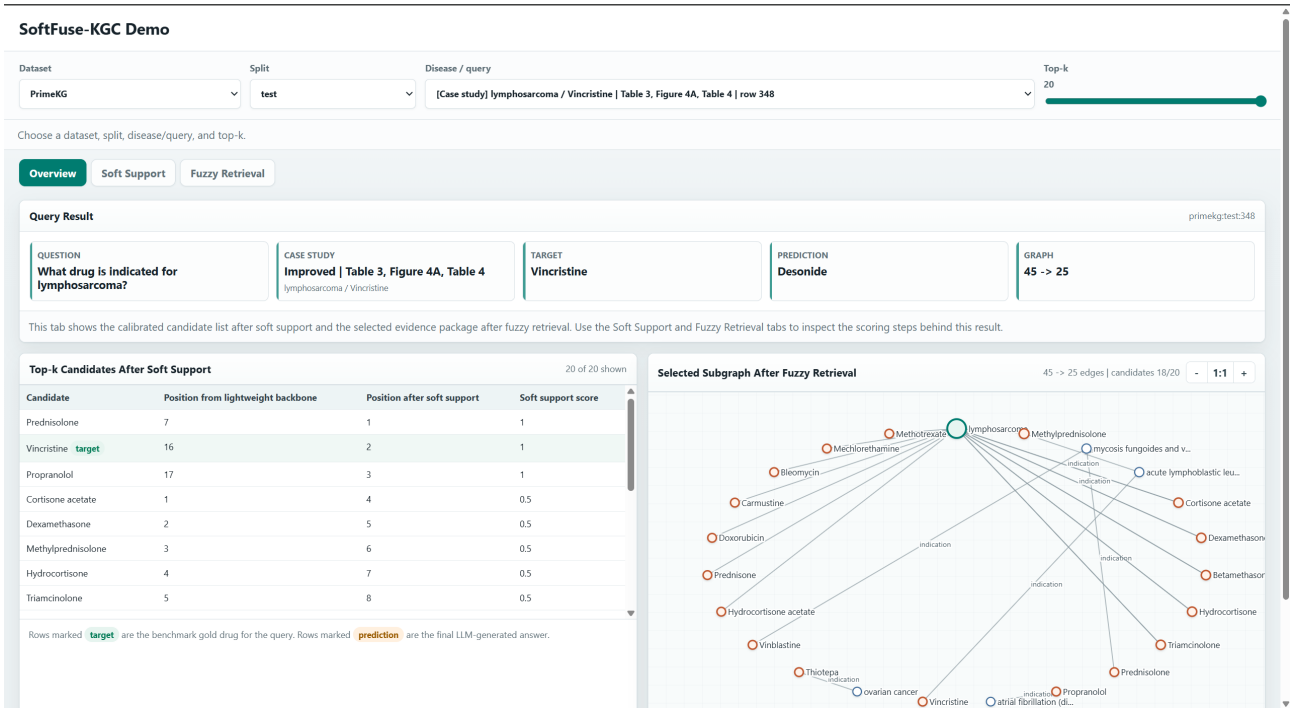


Fig. S1 Overview tab. Controls select the dataset, evaluation split, disease or query case, and displayed top-k. The Query Result panel reports the question, case-study label when available, gold target, prediction, and graph-compression size. The candidate table gives backbone rank, calibrated rank, and soft-support score, with text markers for the target and prediction. The graph panel shows the selected fuzzy evidence subgraph and provides zoom, reset, and drag controls

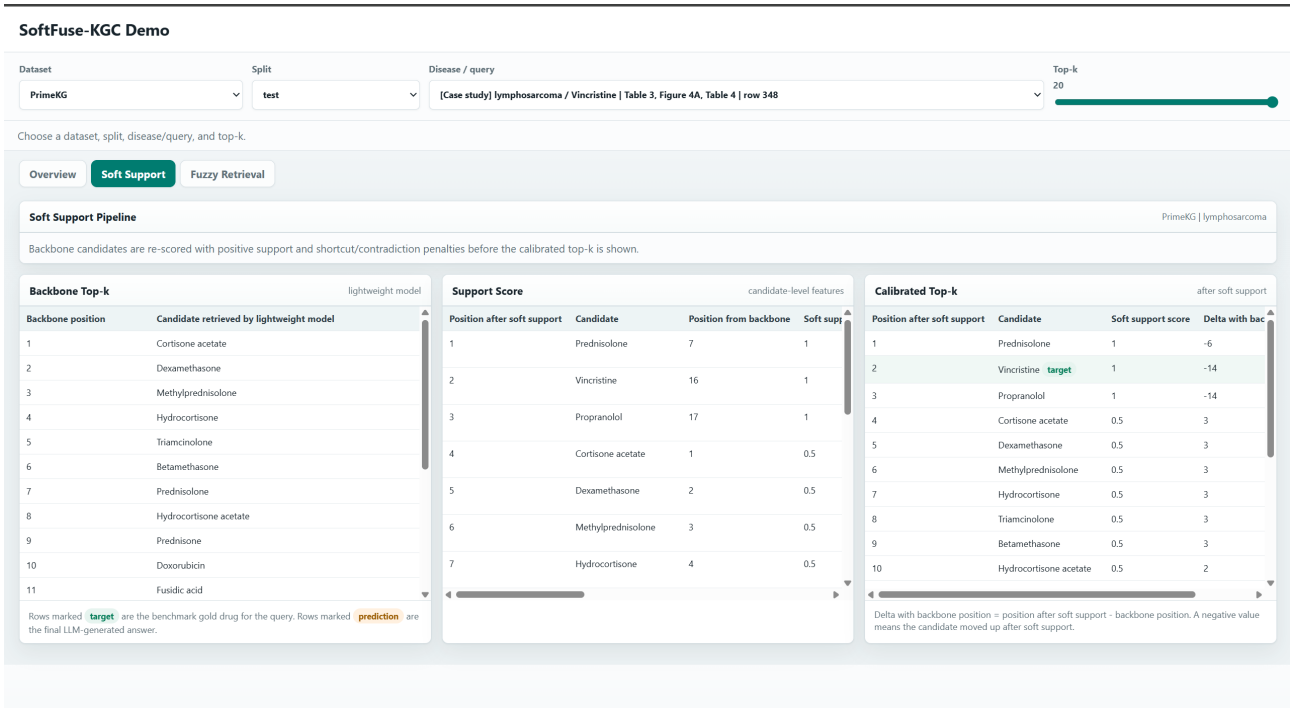


Fig. S2 Soft Support tab. The left panel reports the original lightweight-backbone top-k order. The center panel shows the final support score and available feature terms, including positive evidence, shared-gene or bridge evidence, and shortcut or contradiction penalties. The right panel gives the calibrated top-k. "Delta with backbone position" is calibrated rank minus backbone rank; negative values indicate upward movement

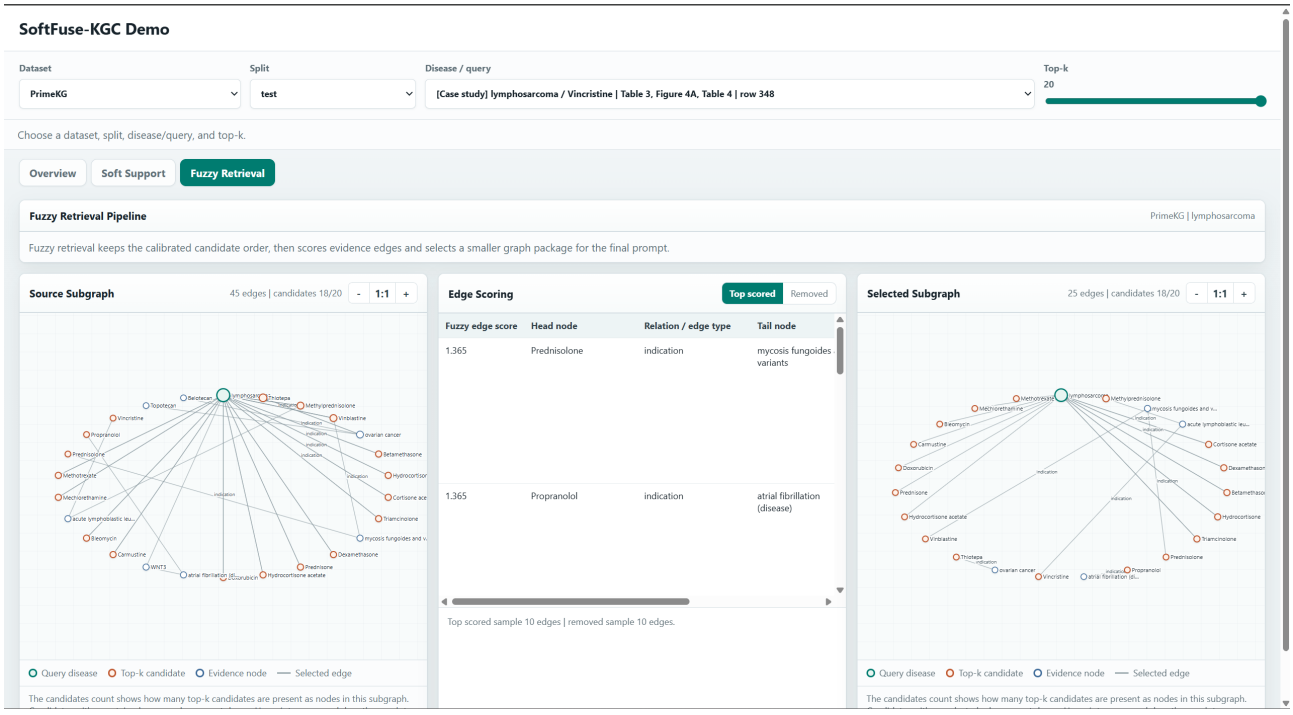


Fig. S3 Fuzzy Retrieval tab. The left panel reports the source subgraph, source-edge count, and candidate coverage. The center panel lists fuzzy edge scores and their component terms and can show either retained or removed evidence. The right panel shows the selected compact subgraph. Labels and the legend identify the query disease, top-k candidates, evidence nodes, and selected edges; color is a redundant visual cue. Zoom, reset, and drag controls support inspection

S1 Soft-Support Feature and Weight Assignment

Equation numbers in this Online Resource refer to the corresponding equations in the main article.

$$
\begin{aligned}
\text{Eq. (2):}\quad E_i(q) &= \omega_{\text{touch}} T_i(q) + \omega_{\text{path}} P_i(q) + \omega_{\text{schema}} A_i(q) + \omega_{\text{density}} \Delta_i(q),\\
\text{Eq. (3):}\quad P_i(q) &= \nu_s M_i^s(q) + \nu_b M_i^b(q)\\
&\quad + \nu_c T_i^c(q) + \nu_q T_i^q(q),\\
M_i^s(q) &= \min\left(1, \frac{|G(c_i) \cap G(q)|}{\eta_s}\right),\\
M_i^b(q) &= \min\left(1, \frac{B_i(q)}{\eta_b}\right),\\
\text{Eq. (5):}\quad \Delta_i(q) &= \min\left(1, \frac{\log(1 + \deg_{S_q^{(0)}}(c_i) + \deg_{S_q^{(0)}}(t_q))}{\eta_\Delta}\right),\\
\text{Eq. (6):}\quad D_i(q) &= \max\left(\mathbb{I}[c_i \leftrightarrow t_q \text{ in } S_q^{(0)}], \min\left(1, \frac{\deg_{\text{direct}}(c_i)}{\eta_D}\right)\right),\\
\text{Eq. (7):}\quad X_i(q) &= \mathbb{I}[c_i \leftrightarrow t_q \text{ through contradiction-like evidence}],\\
\text{Eq. (8):}\quad \widetilde{S}_{\theta_d}(c_i, q) &= \alpha_{E,d} E_i(q) - \alpha_{D,d} D_i(q) - \alpha_{X,d} X_i(q),\\
S_{\theta_d}(c_i, q) &= \mu_d R_i(q) + (1 - \mu_d) N_q(\widetilde{S}_{\theta_d}(c_i, q)).
\end{aligned}
$$

When $\mu_d > 0$, the within-query normalizer in Eq. (8) is

$$
N_q(z_i) = \begin{cases} \dfrac{z_i - \min_j z_j}{\max_j z_j - \min_j z_j}, & \max_j z_j > \min_j z_j,\\ 0.5, & \max_j z_j = \min_j z_j. \end{cases}
$$

S1.1 Feature Selection Method

The soft-support feature family was obtained through an audit-and-calibration procedure. First, for each validation query and each top-20 candidate, a candidate-level feature table was built from the retrieved source subgraph and the training evidence graph. The initial table included type/schema flags, ontology-support flags, candidate evidence contact, query evidence contact, direct candidate-query contact, contradiction or conflict flags, and local count features. For mechanistic datasets, the same procedure was extended to compound-gene, disease-gene, and gene-gene neighborhoods, producing shared-gene, bridge, candidate-touch, and query-touch counts.

The candidate attributes were then summarized by frequency, value range, and gold-versus-non-gold separation. Attributes that were constant or nearly constant on the candidate set were not used as active ranking signals. Attributes that were too sparse but had a clear negative or diagnostic interpretation were retained only as small penalty channels. Attributes that appeared often enough and had an interpretable biomedical role were grouped into the general formula: $T_i(q)$ for evidence contact, $P_i(q)$ for mechanism-oriented support, $A_i(q)$ for schema compatibility when informative, $\Delta_i(q)$ for bounded local density, $D_i(q)$ for direct shortcut or hub-like evidence, and $X_i(q)$ for contradiction-like evidence.

Candidate-stage formulas were then probed as simple score variants before freezing the final coefficient family. The validation rule was to preserve Gold@20, improve or maintain MRR@20, and reduce shortcut exposure. This procedure is why the final soft-support formula separates positive evidence from direct-link penalties instead of using a hard ontology filter or a raw count score. On PrimeKG, for example, the selected contact-based b050 row improved MRR@20 to 0.0976, had no worsened queries relative to the backbone, and reduced the top-five direct-link rate from 0.2408 to 0.0072. The count-like capped-evidence contrast dropped to $\text{MRR@20} = 0.0545$ and restored a shortcut-heavy top-five direct-link rate of 0.2452.

The candidate-density feature in Eq. (5) is retained in the general formulation because it was part of this audited feature family, but it is inactive in the reported candidate reranking rows. In the validation audit, local density

behaved as a hub-sensitive count signal at the candidate stage rather than as a reliable therapeutic-support signal. It is therefore assigned zero candidate-stage weight and retained later only as a small edge-level tie breaker in Eq. (13).

S1.2 Weight Assignment Method

Weights and normalizers were fixed using validation data only. Candidate-stage settings were retained when they preserved Gold@20 and improved or maintained MRR@20 while reducing shortcut exposure. For Eq. (3), η_s and η_b are saturation thresholds for shared-gene and bridge counts, and $\nu_s, \nu_b, \nu_c, \nu_q$ distribute positive support across shared-gene, bridge, candidate-touch, and query-touch channels. For Hetionet and DRKG, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$.

For Eq. (6), η_D is the 95th percentile of nonzero compound-to-disease direct degrees when direct-degree statistics are used. This high-percentile normalizer penalizes generic direct-disease hubs without letting a single extreme hub define the scale. Otherwise, the direct edge flag alone is used. For Eq. (8), N_q is the identity map when $\mu = 0$; when $\mu > 0$, the min–max rule above is applied over the 20 candidates for the same query.

The coefficients below describe the original support formulation. Section S12 adds automatic native-formula selection, shared-feature learned calibration, and coefficient transfer without target calibration labels.

S1.3 Frozen Assigned Values

Table S1 reports the frozen feature instantiations, weights, and normalizers.

Table S1 Dataset-specific soft-support coefficients and feature instantiations. These values instantiate Eq. (2), Eq. (3), Eq. (5), Eq. (6), Eq. (7), and Eq. (8).

Dataset	Positive support instantiation	Penalty and rank-prior coefficients	Normalizers and inactive terms
PrimeKG	In Eq. (2), $E_i(q)$ is contact-based: $E_i(q) = 1$ when the candidate has positive evidence contact in the source subgraph and 0 otherwise.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (6), $D_i(q)$ is the direct candidate-query flag; in Eq. (7), $X_i(q)$ uses contradiction-like evidence when available.	The path term in Eq. (3), density term in Eq. (5), and direct-degree normalizer in Eq. (6) are inactive in the final candidate score.
PharmKG	In Eq. (2), $E_i(q)$ is the evidence-positive support flag for the therapeutic-association proxy relation.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (7), $X_i(q)$ is fixed to zero because no stable contradiction-like relation is used.	The path term in Eq. (3), density term in Eq. (5), and direct-degree normalizer in Eq. (6) are inactive in the final candidate score.
Hetionet	In Eq. (3), mechanistic support combines shared genes, gene-gene bridge evidence, candidate touch, and query touch. Candidate-gene cap $= 120$, disease-gene cap $= 300$, bridge cap $= 30$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (6), direct shortcut evidence includes direct drug-disease evidence and generic direct-disease degree.	In Eq. (3), $\eta_s = 2$, $\eta_b = 5$, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$. In Eq. (6), $\eta_D = 5.0$. The candidate-density term in Eq. (5) is inactive.
DRKG	In Eq. (3), mechanistic support combines shared genes, gene-gene bridge evidence, candidate touch, and query touch. Candidate-gene cap $= 120$, disease-gene cap $= 240$, bridge cap $= 40$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.10$, $\alpha_X = 0.10$, and $\mu = 0.80$. The rank prior is retained because support features are sparse and hub-sensitive.	In Eq. (3), $\eta_s = 2$, $\eta_b = 6$, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$. In Eq. (6), $\eta_D = 52.95$. The candidate-density term in Eq. (5) is inactive.

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Table S1 (continued)

Dataset	Positive support instantiation	Penalty and rank-prior coefficients	Normalizers and inactive terms
repoDB	In Eq. (2), diagnostic support combines approved direct evidence, compound-gene evidence, and gene-bridge evidence: 0.55 approved direct, $0.30\,\min(1, \text{compound gene}/8)$, and $0.15\,\min(1, \text{gene bridge}/12)$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0$, $\alpha_X = 0.20$, and $\mu = 0.95$. The high rank-prior mixture reflects sparse query-specific mechanistic support in the local disease namespace.	The direct-degree feature in Eq. (6) is computed but inactive because $\alpha_D = 0$. The candidate-density term in Eq. (5) is inactive. In Eq. (7), the failed/diagnostic-like penalty is retained as $X_i(q)$.

S2 Fuzzy Edge-Scoring Feature and Weight Assignment

$$\text{Eq. (10):} \quad B(e,q) = \max_{c_i \in C_q^{\text{in}}: e \sim c_i} w_b(r_i),$$

$$w_b(r_i) = \begin{cases} 1.00, & 1 \le r_i \le 5, \\ 0.60, & 6 \le r_i \le 10, \\ 0.25, & 11 \le r_i \le 20, \end{cases}$$

$$\begin{aligned} \text{Eq. (11):} \quad S_{\text{edge}}(e,q) &= \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\ &\quad + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\ &\quad - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned}$$

$$\begin{aligned} \text{Eq. (12):} \quad T_e^c(q) &= \mathbb{I}[e \text{ touches at least one candidate}], \\ T_e^q(q) &= \mathbb{I}[e \text{ touches the query entity}], \\ H_e(q) &= \mathbb{I}[e \text{ touches a top-band candidate}], \end{aligned}$$

$$\text{Eq. (13):} \quad \rho_e(q) = \frac{\log(1 + \deg_{S_q^{(0)}}(u)) + \log(1 + \deg_{S_q^{(0)}}(v))}{\eta_\rho}.$$

S2.1 Feature Selection Method

The fuzzy-retrieval feature family was selected after the candidate order had been fixed. The first step was to convert each retrieved source subgraph into a triple-level feature table. For each evidence triple, the audit recorded whether it touched a candidate, which candidate rank band it touched, whether it touched a top-band candidate, whether it touched the query entity, whether it formed a direct candidate-query shortcut, whether it inherited a contradiction-like flag, its relation family when relation labels were stable, and a bounded local-density hint.

These triple attributes were summarized by occurrence frequency and by their effect on coverage-preserving graph compression. Features were retained when they controlled one of the retrieval decisions needed by the fuzzy package: preserving candidate and query anchoring, transferring confidence from the fixed candidate ranking, prioritizing interpretable biomedical relation families, demoting direct or negative-like shortcut evidence, or breaking ties among otherwise comparable triples. This audit produced the candidate-band score $B(e,q)$, incidence indicators $T_e^c(q)$ and $T_e^q(q)$, the top-candidate indicator $H_e(q)$, relation-family priority $F_r(e)$, shortcut and contradiction penalties $D_e(q), X_e(q)$, and the density term $\rho_e(q)$.

The resulting features were then tested in coverage-then-fill retrieval variants. A feature combination was kept only if the selected graph preserved the candidate order, Gold@20, candidate MRR@20, and candidate coverage while reducing the average evidence package or lowering shortcut-heavy evidence. On the PrimeKG validation audit, the selected Fuzzy-compact setting preserved the soft-support ranking metrics and candidate coverage while reducing the average subgraph from 60.45 to 32.56 triples. The intermediate Fuzzy-balanced and Fuzzy-shortcut variants retained 36.714 triples on average. All fuzzy variants reduced direct-shortcut exposure. This is why fuzzy retrieval is treated as evidence packaging rather than as a second candidate reranker.

S2.2 Weight Assignment Method

Fuzzy edge weights were fixed on validation data after the input candidate order had already been selected. Eligible settings had to preserve Gold@20 and candidate MRR@20, maintain candidate coverage, and reduce or clean the evidence graph. Candidate rank bands are assigned after the top-20 list is fixed: ranks 1–5 receive weight 1.00, ranks 6–10 receive weight 0.60, and ranks 11–20 receive weight 0.25. If an edge touches multiple candidates, Eq. (10) uses the highest touched-candidate band weight; if it touches no candidate and is not part of a candidate-anchored path, its band score is zero.

The density term in Eq. (13) uses $\eta_\rho = 10$ for PrimeKG, Hetionet, DRKG, and repoDB. PrimeKG uses a capped local-density hint, whereas Hetionet, DRKG, and repoDB use the log-degree density term shown above. PharmKG keeps its native local touch-count density hint. The density coefficient is 0.05 in the selected fuzzy rows, so density acts as a tie-breaking signal after candidate coverage, relation-family evidence, and penalty terms have been applied.

S2.3 Frozen Assigned Values

Table S2 reports the frozen edge-scoring weights.

$$\text{Eq. (10):}\quad B(e,q) = \max_{c_i \in C_q^{\text{in}} : e \sim c_i} w_b(r_i),$$

$$w_b(r_i) = \begin{cases} 1.00, & 1 \le r_i \le 5, \\ 0.60, & 6 \le r_i \le 10, \\ 0.25, & 11 \le r_i \le 20, \end{cases}$$

$$\begin{aligned} \text{Eq. (11):}\quad S_{\text{edge}}(e,q) &= \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\ &\quad + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\ &\quad - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned}$$

$$\begin{aligned} \text{Eq. (12):}\quad T_e^c(q) &= \mathbb{I}[e \text{ touches at least one candidate}], \\ T_e^q(q) &= \mathbb{I}[e \text{ touches the query entity}], \\ H_e(q) &= \mathbb{I}[e \text{ touches a top-band candidate}], \end{aligned}$$

$$\text{Eq. (13):}\quad \rho_e(q) = \frac{\log(1 + \deg_{S_q^{(0)}}(u)) + \log(1 + \deg_{S_q^{(0)}}(v))}{\eta_\rho}.$$

Table S2 Frozen fuzzy edge-scoring weights. These values instantiate Eq. (10), Eq. (11), Eq. (12), and Eq. (13). All datasets use rank bands 1–5, 6–10, and 11–20 with weights 1.0, 0.6, and 0.25, respectively.

Dataset	Candidate, query, and density weights	Relation-family priorities	Penalty terms
PrimeKG	In Eq. (10)–Eq. (11), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.05$, $\lambda_{\text{top}} = 0$, and $\lambda_\rho = 0.05$. In Eq. (13), density uses a capped local-density hint with $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 0$; no relation-family priority is used.	In Eq. (11), the direct shortcut penalty is $\lambda_d = 0.80$, and the contradiction penalty is $\lambda_x = 0.10$.
PharmKG	In Eq. (10)–Eq. (11), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.05$, $\lambda_{\text{top}} = 0$, and $\lambda_\rho = 0.05$. The density term keeps the native local touch-count hint.	In Eq. (11), $\lambda_f = 0$; no relation-family priority is used.	In Eq. (11), the direct shortcut penalty is $\lambda_d = 0.80$, and the contradiction penalty is $\lambda_x = 0$.
Hetionet	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is compound-gene 0.30, disease-gene 0.30, gene-gene 0.20, direct drug-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.50$ and $\lambda_x = 0$.
DRKG	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is compound-gene 0.30, disease-gene 0.30, gene-gene 0.20, target-treats 0.05, auxiliary compound-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.50$ and $\lambda_x = 0$.
repoDB	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is target-approved 0.10, failed-diagnostic 0.02, compound-gene 0.35, gene-gene 0.25, other compound-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.40$ and $\lambda_x = 0.15$ for failed-diagnostic evidence.

S3 Fuzzy Retrieval Budgets

$$S_q^{\text{fuzzy}} = \text{TopEdges}\Big(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\Big), \quad \text{Eq. (14)}$$

$$C_q^{\text{fuzzy}} = C_q^{\text{in}}. \quad \text{Eq. (15)}$$

The budget values in Table S3 instantiate Eq. (14), while Eq. (15) fixes the candidate order after fuzzy graph selection.

Table S3 Frozen fuzzy retrieval budgets and coverage checks. These settings instantiate the graph budget $\tau_{d,q}$ in Eq. (14); candidate rankings are preserved as in Eq. (15).

Dataset	Budget rule	Selection passes	Frozen coverage result
PrimeKG	In Eq. (14), retain ratio 0.55, minimum 18, maximum equal to source size.	Candidate coverage pass, then global fill by edge score.	Validation 60.45 → 32.56 triples; test 59.93 → 32.34 triples. Candidate coverage preserved: 1.0.
PharmKG	In Eq. (14), retain ratio 0.55, minimum 18, maximum equal to source size.	Candidate coverage pass, then global fill by edge score.	Validation and test 100 → 55 triples. Candidate, top-band, middle-band, and tail-band coverage preserved: 1.0.
Hetionet	In Eq. (14), retain ratio 0.55, minimum 28, maximum 48.	Candidate coverage pass, query coverage pass, then global fill.	Validation and test 80 → 44 triples. Candidate coverage preserved: 1.0.
DRKG	In Eq. (14), retain ratio 0.55, minimum 45, maximum 60, family minimum 3.	Candidate coverage pass, query coverage pass, family preservation pass, then global fill.	Validation and test 100 → 55 triples. Candidate coverage preserved: 1.0.
repoDB	In Eq. (14), retain ratio 0.55, minimum 45, maximum 60. Family minimums are compound-gene 8, gene-gene 6, target 1, and failed 0.	Candidate coverage pass, query coverage pass, family preservation pass, then global fill.	Validation and test 100 → 55 triples. Candidate coverage preserved: 1.0.

S4 Algorithm

$$S_{\theta_d}(c_i, q) = \mu_d R_i(q) + (1-\mu_d) N_q(\widetilde{S}_{\theta_d}(c_i, q)), \quad \text{Eq. (8)}$$

$$c_i \prec c_j \iff S_i > S_j \text{ or } (S_i = S_j \text{ and } r_i^{(0)} < r_j^{(0)}), \quad \text{Eq. (9)}$$

$$\begin{aligned} S_{\text{edge}}(e, q) = {} & \lambda_f F_r(e) + \lambda_b B(e, q), \\ & + \lambda_c T_e^c(q) + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) \\ & - \lambda_d D_e(q) - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned} \quad \text{Eq. (11)}$$

$$S_q^{\text{fuzzy}} = \text{TopEdges}\Big(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\Big), \quad \text{Eq. (14)}$$

$$\mathbf{z}_e = \text{Adapter}\left([\mathbf{h}_e^{\text{global}}; \mathbf{h}_e^{\text{local}}]\right). \quad \text{Eq. (16)}$$

Algorithm S1 summarizes the complete inference sequence for one query.

Algorithm S1 SoftFuse-KGC inference for one query

Require: Query $q = (?, r_q, t_q)$, candidate universe $\mathcal{C}_d$, graph-compatible ranker, source graph retriever, support features, edge score S_{edge}

Retrieve the initial top-20 candidate list $C_q^{(0)}$.
Build the source evidence subgraph $S_q^{(0)}$ from the query, candidates, and available rule/path evidence.
Compute candidate support scores $S_{\theta_d}(c_i, q)$ using Eq. (8).
Re-rank candidates with the tie-breaking rule in Eq. (9) to obtain C_q^{soft}.
Assign candidate bands from C_q^{soft} and score evidence triples with Eq. (11).
Select S_q^{fuzzy} using the coverage and fill passes in Table S3.
Compute graph-enhanced query and candidate embeddings with the graph adapter in Eq. (16).
Insert `[QUERY]` and `[ENTITY]` embeddings into the candidate-constrained LLM prompt.
Generate the final answer from the calibrated candidate list.
return Prediction $\hat{y}_q$.

S5 Dataset Implementation Details

The end-to-end implementation uses a reproduced DrKGC-style graph-LLM setup: a graph-compatible R-GCN source provides candidate identifiers, local subgraphs, and structural embeddings for the graph adapter; the LLM is fine-tuned with LoRA; validation and test use $K = 20$ candidates. The primary dataset runs use Llama-3.2-3B with LoRA $r = 32$, $\alpha = 32$, dropout 0.1, learning rate 2×10^{-4}, source/target lengths $768/64$, deterministic one-beam decoding without sampling, and a single NVIDIA RTX 4090 GPU with 24GB memory. Table S4 reports the dataset-specific graph sources and frozen generation settings. These graph-adapter pipeline sources are distinct from the separately trained structure-only baselines reported later; their embedding dimensions therefore need not match.

Table S4 Graph-LLM implementation and decoding settings. Common LoRA, decoding, and hardware settings are stated in the paragraph above.

Dataset	Graph source	Training detail	Frozen decoding and graph size
PrimeKG	R-GCN; 4 rel.; 10453×256 embeddings.	Batch 1; grad. acc. 8.	New 16; rep. 1.00; no-repeat 0; graph $59.93 \rightarrow 32.34$.
PharmKG	R-GCN; 28 rel.; 7247×128 embeddings.	Batch 1; grad. acc. 16.	New 16; rep. 1.05; no-repeat 0; graph $100 \rightarrow 55$.
Hetionet	R-GCN-compatible; 24 rel.; 47031×128 embeddings.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $80 \rightarrow 44$.
DRKG	R-GCN consistency source; 85 rel.; R-GCN-ready IDs.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $100 \rightarrow 55$.
repoDB	R-GCN; 63 rel.; readable names plus canonical IDs.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $100 \rightarrow 55$.

Table S5 lists the question templates and evidence-path retrieval packages used for each dataset.

Table S5 Pre-specified question templates and evidence-path retrieval rules. Task templates and retrieval rules format candidate-constrained prompts and package local evidence.

Dataset	Question template	Path-rule or retrieval package
PrimeKG	*What drug is indicated for {disease}?*	Drug–target–gene–disease and drug–target–gene–ppi–gene–disease paths; contraindication-like drug–disease evidence is retained as a negative/safety cue.
PharmKG	*What drug is therapeutically associated with {disease}?*	The T relation is the therapeutic-association proxy; retrieval uses incident-edge and shortest-path fallback evidence.
Hetionet	*What compound treats {disease}?*	CbG–DaG, CdG–DdG, CuG–DuG, one-hop GiG bridges, and direct CpD evidence.
DRKG	*What compound treats {disease}?*	Compound:Gene–Gene:Disease, direct Compound:Disease, and Gene:Gene bridge evidence for the DrugBank treats head-prediction target.
repoDB	*What drug is approved for {disease}?*	Compound:Gene–Gene:Gene, Compound:Disease, and failed-diagnostic evidence retained as diagnostic rather than positive approved-indication context.

S5.1 Prompt and Output Example

All datasets use the same candidate-constrained prompt pattern. The following PrimeKG example illustrates the input schema and supervised output:

> You are a biomedical scientist. The task is to predict the answer based on the given question, and you only need to answer one entity. The answer must be in ('Fusidic acid', 'Cortisone acetate', 'Hydrocortisone', 'Betamethasone', 'Tetracycline', 'Ofloxacin', 'Ciprofloxacin', 'Demeclocycline', 'Dexamethasone', 'Triamcinolone', 'Prednisolone', 'Methylprednisolone', 'Hydrocortisone acetate', 'Methdilazine', 'Norfloxacin', 'Doxycycline', 'Prednisone', 'Alimemazine', 'Doxylamine', 'Phenylephrine').
> You can refer to the entity embeddings: 'allergic rhinitis': `[QUERY]`, 'Fusidic acid': `[ENTITY]`, 'Cortisone acetate': `[ENTITY]`, 'Hydrocortisone': `[ENTITY]`, ..., 'Phenylephrine': `[ENTITY]`.
> Question: What drug is indicated for allergic rhinitis?
> Answer:

The corresponding supervised output is *Hydrocortisone*. During evaluation, the generated string is normalized and matched against the supplied candidate list; unmatched strings are counted as invalid predictions.

S6 PrimeKG Variant Formulas

PrimeKG is used here as the representative formula summary.

S6.1 Soft-Support Variants

$$
\begin{aligned}
\text{Selected family:}\quad & \widetilde{S}(c_i, q) = \alpha_E E_i(q) - \alpha_D D_i(q) - \alpha_X X_i(q), \\
\text{b025}: \quad & E_i(q) - 0.25 D_i(q) - 0.05 X_i(q), \\
\text{b050}: \quad & E_i(q) - 0.50 D_i(q) - 0.10 X_i(q), \\
\text{bcap}: \quad & \underbrace{\min(E_i^{\text{count}}(q), 2)/2}_{\text{normalized capped support}} - 0.50 D_i(q) - 0.10 X_i(q).
\end{aligned}
$$

The selected PrimeKG contact-support family is a special case of Eq. (8) with $\mu = 0$. The final row uses b050. The b025 row tied b050 in validation, but b050 was retained because it applies the stronger shortcut penalty while preserving the same candidate ranking. The capped-evidence contrast underperformed because it reintroduced shortcut-heavy candidates.

S6.2 Fuzzy-Retrieval Variants

$$S_{\text{edge}}(e,q) = B(e,q) + 0.35T_e^c(q) + 0.05T_e^q(q) - \lambda_d D_e(q) - 0.10X_e(q) + 0.05\rho_e(q).$$

The PrimeKG fuzzy edge score uses Eq. (11) with no relation-family term. The selected Fuzzy-compact variant (called *tight* in the configuration audit) sets $\lambda_d = 0.80$, retain ratio 0.55, and minimum keep 18. The Fuzzy-shortcut contrast (called *directplus* in the audit) sets $\lambda_d = 1.00$, retain ratio 0.67, and minimum keep 20. Both preserve candidate ranking, but Fuzzy-compact gives the smaller and cleaner graph package on validation. Table S6 summarizes the selected and contrastive PrimeKG rows.

Table S6 PrimeKG variant-selection summary. Candidate metrics are validation metrics for soft support and held-out test metrics for the locked ablation rows.

Row	Gold@20	MRR@20	Hits@1	Hits@10	Selection note
Backbone raw, validation	0.202	0.0538	0.024	0.158	Reference candidate order.
Ontology hard filter, validation	0.014	0.0039	0.002	0.012	Negative control; brittle under missing evidence.
Soft support b050, validation	0.202	0.0976	0.056	0.182	Selected soft-support row.
Capped-evidence contrast, validation	0.202	0.0545	0.022	0.158	Retained only as a contrast row.
Backbone raw, test	0.240	0.0646	0.024	0.192	Locked test reference.
Soft support, test	0.240	0.1253	0.072	0.222	Candidate-stage gain.
Fuzzy retrieval, test	0.240	0.1253	0.072	0.222	Candidate order preserved; graph $59.93 \rightarrow 32.34$.

S7 LLM Model Comparison

This section retains the original generator and decoding-configuration comparisons. Generated strings are matched against the supplied candidates. Invalid answers are not exact matches, but an adjusted-rank metric can still credit a present gold candidate; adjusted MRR should therefore not be read as generation accuracy. The current standardized module and intervention comparisons are reported in the main text and Sections S12–S13, separately from these earlier model-selection diagnostics.

Table S7 reports the PrimeKG generator comparison.

Table S7 PrimeKG LLM comparison on the fuzzy retrieval row. All rows use the same candidate list and fuzzy evidence package. The configuration column reports max new tokens / repetition penalty / no-repeat n-gram size.

Model	Config	E2E MRR@20	E2E Hits@10	Pred. in cand.	Invalid	Top1 copy	Interpretation
Llama-3.2-3B	16/1.00/0	0.0747	0.218	0.998	0.002	0.010	Primary generator; high candidate validity without top-ranked candidate copying.
Llama-3-8B	16/1.00/0	0.1253	0.222	1.000	0.000	1.000	Diagnostic generator behavior; all test outputs follow the first candidate.
MedLlama-3-8B	16/1.00/0	0.1253	0.222	1.000	0.000	1.000	Same candidate-order-following behavior as Llama-3-8B.

Table S8 reports the corresponding PharmKG comparison.

Table S8 PharmKG LLM comparison on the fuzzy retrieval row. The first three rows use the frozen PharmKG decoding configuration. Mistral-7B is included as a diagnostic comparison.

Model	Config	E2E MRR@20	E2E Hits@10	Pred. in cand.	Invalid	Top1 copy	Interpretation
Llama-3.2-3B	16/1.05/0	0.0210	0.062	0.412	0.588	0.412	Most stable PharmKG generator among the frozen rows.
Llama-3-8B	16/1.05/0	0.0203	0.064	0.274	0.726	0.274	Lower candidate validity; no PrimeKG-like top1-copy collapse.
MedLlama-3-8B	16/1.05/0	0.0186	0.060	0.022	0.978	0.022	Mostly invalid under the candidate-constrained output rule.
Mistral-7B	32/1.00/0	0.0186	0.060	0.014	0.986	0.000	Diagnostic comparison; candidate validity remains very low.

Table S8 shows a different failure mode from the PrimeKG 8B rows. Llama-3.2-3B gives the highest candidate-constrained validity on PharmKG, although invalid outputs remain frequent because the transfer task uses a noisier therapeutic-association proxy. Llama-3-8B has slightly higher Hits@10 but lower candidate validity, while MedLlama-3-8B and Mistral-7B mostly generate strings outside the supplied candidate list.

S7.1 Decoding-Configuration Check

The deterministic decoding sweep used for configuration selection varied max new tokens, repetition penalty, and no-repeat n-gram size. The sweep was performed on the validation split with the primary generator, and the selected configuration was then held fixed for the model comparison. This avoids selecting a separate decoding policy for each larger model after observing comparison outcomes. Table S9 therefore separates two checks: the full eight-configuration sweep showing that the frozen decoding setting itself does not induce top1-copy behavior in Llama-3.2-3B, and the larger-model frozen checks showing that top1-copy appears when the generator is changed to Llama-3-8B or MedLlama-3-8B under the same selected configuration.

Table S9 PrimeKG decoding-configuration check for top1-copy behavior. Top1-copy means that the generated entity is the first candidate in the supplied list. The Llama-3.2-3B rows report the complete validation sweep; the 8B rows report the frozen configuration used in the model comparison.

Model/protocol	Config	New	Rep.	No-rep.	Val. E2E MRR@20 / Hits@10	Val. top1 copy	Test top1 copy	Interpretation
Llama-3.2-3B sweep	cfg01	16	1.00	0	0.0585/0.174	0.008	0.010	Frozen setting selected because all swept settings tied on the fuzzy row.
Llama-3.2-3B sweep	cfg02	16	1.05	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg03	16	1.10	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg04	16	1.10	3	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg05	24	1.00	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg06	24	1.05	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg07	24	1.10	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg08	24	1.10	3	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3-8B frozen check	cfg01	16	1.00	0	0.0976/0.182	1.000	1.000	Test outputs: 464 non-gold top1 copies and 36 exact-target top1 cases.
MedLlama-3-8B frozen check	cfg01	16	1.00	0	0.0976/0.182	1.000	1.000	Same validation and test breakdown as Llama-3-8B.

The PrimeKG sweep shows that changing deterministic decoding length, repetition penalty, and no-repeat n-gram size did not change the primary generator's fuzzy-row validation metrics or create top1-copy collapse. The collapse appears only in the larger-model frozen checks, where both 8B generators copy the first candidate for every validation and test example. Their higher adjusted E2E MRR@20 is therefore interpreted as candidate-order following after the soft-support/fuzzy candidate list has already been supplied, not as stronger use of the evidence text or graph package.

S8 Ablation, Sensitivity, and Robustness

Table S10 isolates the candidate-reranking and evidence-packaging contributions on the PrimeKG test split.

Table S10 PrimeKG held-out test ablation. The soft-support row changes candidate ordering; fuzzy retrieval changes evidence packaging while preserving the candidate order.

Row	Gold@20	MRR@20	Hits@1	Hits@10	Avg. graph	Note
Backbone raw	0.240	0.0646	0.024	0.192	59.93	Initial graph-compatible source.
Soft support	0.240	0.1253	0.072	0.222	59.93	Candidate $\Delta\mathrm{MRR@20} = +0.0608$.
Fuzzy retrieval	0.240	0.1253	0.072	0.222	32.34	Ranking unchanged; graph reduced by 27.59 triples on average.

Table S11 evaluates retrieval-rule and question-template sensitivity with a fixed candidate order.

Table S11 PrimeKG rule-package and question-template sensitivity. Candidate order is fixed in these checks.

Variant	Candidate MRR@20	E2E MRR@20	E2E Hits@10	Invalid	Avg. graph	Note
Main rules	0.1253	0.0747	0.218	0.002	32.34	Canonical setting.
No rules	0.1253	0.0747	0.218	0.020	59.93	Similar rank metric but larger graph and higher invalid rate.
Random rules	0.1253	0.0747	0.218	0.002	32.34	Diagnostic control; not selected over main rules.
Canonical question	0.1253	0.0747	0.218	0.002	32.34	Baseline wording.
Treatment wording	0.1253	0.0747	0.218	0.000	32.34	Prediction string changes in 3.0% of test examples.
Medication wording	0.1253	0.0747	0.218	0.002	32.34	Prediction string changes in 0.6% of test examples.
Association-neutral wording	0.1253	0.0747	0.218	0.000	32.34	Prediction string changes in 4.6% of test examples.

Table S12 reports small score perturbations and edge-drop robustness checks.

Table S12 PrimeKG small-noise robustness on the held-out test split. Score-noise variants perturb support scores before reranking; edge-drop variants remove approximately 5% of selected subgraph edges.

Variant	MRR@20	Hits@10	Same top1	Rank change	Avg. graph
No noise	0.1253	0.222	1.000	0.000	32.34
Score noise s1	0.0779	0.220	0.086	0.206	32.34
Score noise s2	0.0779	0.216	0.096	0.214	32.34
Score noise s3	0.0763	0.220	0.086	0.214	32.34
Edge drop s1	0.1253	0.222	1.000	0.000	30.49
Edge drop s2	0.1253	0.222	1.000	0.000	30.49
Edge drop s3	0.1253	0.222	1.000	0.000	30.49

The sensitivity checks show that the PrimeKG result is not driven by one question wording. The robustness check separates two effects: edge dropout leaves the fixed candidate ranking unchanged, whereas support-score perturbation changes close score ties and reduces candidate MRR. Fixed-order ranking invariance alone does not establish retained graph information; Section S13 measures this directly. This highlights a calibration limitation of the current support score.

S9 Fuzzy Encoder Probe

The fuzzy encoder probe tests whether fuzzy evidence weights should enter message passing directly rather than only the retrieval stage. It remains diagnostic because it worsens more validation cases than it improves. Table S13 reports the validation comparison.

Table S13 PrimeKG fuzzy encoder probe on validation. The probe uses the same validation split as the retrieval-selection comparison.

Row	Gold@20	MRR@20	Hits@1	Hits@3	Hits@10	Probe note
Soft support	0.202	0.0976	0.056	0.134	0.182	Candidate-stage reference.
Fuzzy retrieval	0.202	0.0976	0.056	0.134	0.182	Candidate order preserved; avg. graph 32.56.
Fuzzy encoder probe	0.202	0.0537	0.012	0.058	0.176	Improves 8 queries, worsens 70, and leaves 422 unchanged relative to fuzzy retrieval.

This probe does not support replacing the retrieval-stage design with the tested weighted-message-passing formulation. The separate structural and frozen-model comparisons in Section S13 characterize evidence selection more directly.

S10 Structure-Only Baseline Training Settings

Table S14 reports the training settings for all structure-only baselines.

Table S14 Structure-only baseline training settings. Translation/bilinear baselines include TransE, DistMult, ComplEx, and RotatE. GNN baselines include R-GCN and HRGAT.

Dataset	Translation and bilinear baselines	GNN baselines	Candidate and graph protocol
PrimeKG	TransE, DistMult, ComplEx, RotatE: dimension 128, 80 epochs, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	R-GCN and HRGAT: dimension 128, 40 epochs, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	Drug-only universe: 1,801 candidates; training graph: 136,351 triples; target train pairs: 8,388.
PharmKG	TransE, DistMult, ComplEx, RotatE: dimension 256, 30 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2025.	R-GCN and HRGAT: dimension 128, two graph layers, 40 epochs, 32,768 sampled edges per epoch, dropout 0.1, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	Training heads of relation T: 1,342 candidates; training graph: 386,768 triples and 28 relations.
Hetionet	TransE, DistMult, ComplEx, RotatE: dimension 128, 20 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2026.	R-GCN: dimension 128, two graph layers, 80 epochs. HRGAT: dimension 64, two graph layers, 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Compound universe: 1,552 candidates; training graph: 2,249,997 edges and 24 relations.
DRKG	TransE, DistMult, ComplEx, RotatE: dimension 128, 30 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2027.	R-GCN and HRGAT: dimension 128, two graph layers; R-GCN uses 80 epochs and HRGAT uses 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Training target heads: 1,542 candidates; training graph: 187,962 edges and 85 relations.
repoDB	TransE, DistMult, ComplEx, RotatE: dimension 128, 40 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2028.	R-GCN and HRGAT: dimension 128, two graph layers; R-GCN uses 80 epochs and HRGAT uses 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Approved training heads: 1,519 candidates; training graph: 90,804 edges and 63 relations.

S11 Dataset Preprocessing and Leakage Checks

$$q = (?, r_q, t_q), \qquad (c_q^\star, r_q, t_q) \notin \mathcal{G}_{\text{evidence}}^{\text{train}} \quad \text{for validation and test queries.}$$

All datasets are converted into head-prediction tasks of the form $(?, r_q, \text{disease})$. Validation and test splits are constructed before candidate scoring, and the held-out positive target triples are removed from the training evidence graph. Table S15 summarizes target construction, evidence-graph preparation, and leakage checks.

Table S15 Dataset preprocessing and leakage checks. Coverage means that validation/test gold drugs and query diseases remain represented in the training candidate/query universe. Exact leakage means that held-out target triples are absent from the training evidence graph and prepared evaluation inputs.

Dataset	Target construction and split	Evidence graph construction	Leakage and protocol checks
PrimeKG	Drug-restricted indication head prediction with $8{,}388/500/500$ train/validation/test rows. Candidate universe is drug-only, and $K = 20$ candidates are supplied without validation/test gold injection.	Four-relation enriched graph: indication, target, associated-with, and protein-protein interaction. Evaluation prompts align candidates and evidence by query; supervised targets appear only in training outputs.	Validation/test exact-leak counts are 0. Training rows contain supervised targets by design; validation/test rows use no-injection candidate lists.
PharmKG	Relation T is a therapeutic-association proxy. Coverage filtering, train-overlap removal, forbidden-pair removal, and deduplication give $28{,}960/500/500$ rows.	Training T-heads define $1{,}342$ candidates. The enriched graph has $386{,}768$ triples, $7{,}247$ entities, and 28 relations.	Leakage checks pass: valid-positive-in-train $= 0$, test-positive-in-train $= 0$, valid-test overlap $= 0$, enriched-graph held-out target count $= 0$; type errors $= 0$.
Hetionet	The target relation is CtD, with $555/100/100$ train/validation/test rows. The split enforces that validation/test compounds and diseases remain in the training CtD relation.	Held-out CtD target edges are removed from the enriched graph, which contains $2{,}249{,}997$ training edges over $47{,}031$ entities and 24 relations.	Coverage passes for validation and test. The 200 held-out target edges are removed, and exact-leak count is 0 in the graph, raw evaluation inputs, soft-support inputs, and fuzzy-retrieval inputs.
DRKG	The target relation is DrugBank treats Compound-to-Disease, with $3{,}968/500/500$ train/validation/test rows. Validation/test compounds and diseases remain in the training target-relation universe.	The enriched graph keeps selected compound-gene, target-treats, auxiliary compound-disease, disease-gene, and gene-gene families, yielding $187{,}962$ edges over $13{,}918$ entities and 85 relations.	Coverage passes for validation and test. The $1{,}000$ held-out target edges are skipped during graph construction, and valid/test exact-leak counts are 0 in the graph and prepared evaluation inputs.
repoDB	Approved indication pairs form the target relation. Failed or suspended pairs are retained only as diagnostic evidence after conflict handling. The approved target split is $5{,}677/500/500$.	Diseases are local UMLS nodes; mapped DrugBank compounds reuse DRKG-derived compound-gene and gene-gene evidence. The graph has $90{,}804$ edges, $8{,}818$ entities, and 63 relations.	Coverage and schema checks pass. Validation and test exact-leak counts are 0; all task entities have local graph IDs, which does not imply that each entity is aligned to DRKG (Section S11.1).

S11.1 repoDB normalization and cross-resource entity alignment

Normalization and target definition. The repoDB preparation code cleans HTML and whitespace, converts identifier strings to uppercase, and extracts DrugBank IDs matching `DB` followed by five digits and UMLS CUIs matching `C` followed by seven digits. It reads the structured drug and indication fields, falling back to the corresponding raw text fields when necessary. All 10,800 input rows have valid identifiers. Drugs are represented as `Compound::DBxxxxx` and diseases as `Disease::UMLS:Cxxxxxxx`; names are retained for display rather than used as cross-resource join keys. Approved records define positive targets. Terminated, withdrawn, suspended, or failed records form a separate diagnostic relation, not positive indication labels or assumed mechanistic contraindications. Deduplication by canonical drug–disease pair gives 6,677 approved pairs and 3,281 failed-like pairs. The 21 pairs appearing in both groups retain their approved label and are removed from the diagnostic group, leaving 3,260 diagnostic pairs.

Alignment scope and unmatched entities. The join uses exact canonical DrugBank IDs in the prepared DRKG entity map, which contains 13,918 entities; it is not a mapping against the complete raw DRKG. Across approved and diagnostic pairs, 1,130 of 1,571 unique compounds (71.93%) match this map. Among the 1,519 approved compounds, 1,106 match (72.81%). For each of the 2,051 distinct disease CUIs, the code probes three literal forms: `Disease::UMLS:Cxxxxxxx`, `Disease::Cxxxxxxx`, and the bare CUI. None matches. These

checks test identifier equality only; they do not establish the absence of semantic correspondences. No ontology crosswalk, synonym-based disease alignment, or disease-to-gene identifier conversion is performed. Diseases remain local UMLS nodes, and unmatched compounds are retained rather than excluded from the benchmark. Thus local graph-ID coverage and cross-resource alignment coverage are different quantities.

Reuse of DRKG evidence. The graph starts with training approved-indication edges and conflict-cleaned diagnostic edges. It then scans the prepared DRKG training graph for compound–gene edges incident to repoDB compounds with matching IDs, preserving the original relation, direction, and gene ID. Gene–gene bridges are retained only when both endpoints belong to the gene set touched by those compound–gene edges. No DRKG disease–gene edges are imported through a UMLS disease mapping. Diagnostic edges are capped at 80 per disease and 80 per compound; compound–gene edges at 80 per compound; gene–gene edges at 40 per gene and 150,000 overall. Edges are deduplicated, and the caps are applied in the stored input order. This yields 5,677 training approved edges, 3,222 diagnostic edges, 39,017 compound–gene edges, and 42,888 gene–gene edges: 90,804 triples over 8,818 entities and 63 relations. The drop from 3,260 to 3,222 diagnostic pairs occurs during capped graph construction. All task entities are included in the local ID maps even if they lack imported DRKG evidence.

Splitting and leakage checks. The 6,677 unique approved triples are split into 5,677/500/500 training/validation/test triples with seed 2028. A triple is eligible for holdout only if its compound and disease each retain another approved training edge. The resulting 1,519 training compounds define the candidate universe. The splits are disjoint, every validation/test compound and disease remains in the training target-relation universe, and no held-out approved triple occurs in the constructed evidence graph. Diagnostic edges for held-out positive pairs are also excluded. These are exact-triple and entity-coverage checks, not claims that all semantic overlap between resources has been removed. Evaluation candidate lists are produced without gold injection; canonical and numeric IDs are retained separately from display names in the prepared graph–LLM inputs.

The alignment therefore supplies drug-side molecular evidence and gene bridges while leaving disease-side mechanistic context limited to the local graph. This defines the scope of the repoDB setting: evaluation under partial cross-resource alignment, rather than a fully aligned UMLS–DRKG benchmark.

S12 Automatic Calibration and Coefficient Transfer

This section supplements the main-text calibration results with reproducible fitting rules, all-dataset controls, and generated-answer evaluation. The analysis uses the existing 2,100 validation rows as a calibration pool and the complete 2,100 test rows (500 per dataset except 100 for Hetionet). These are the same benchmark examples across variants, not independent additional samples. Candidate membership is fixed to the original top-20 lists. No model or coefficient is selected using test outcomes.

S12.1 Shared features and learned scoring

For candidate c at one-based original rank r_c and query node q, let $d(c)$ count incident source-edge occurrences and $N(c)$ denote its undirected neighbor set. The common feature vector is

$$\mathbf{x}(c,q) = \left[\frac{21-r_c}{20},\ \mathbb{I}[d(c)>0],\ \min\left(1, \frac{\log(1+d(c))}{\log 101}\right),\right.$$
$$\left.\min\left(1, \frac{\log(1+|N(c)\cap N(q)|)}{\log 21}\right),\ -\mathbb{I}[q\in N(c)]\right].$$

These features use source-subgraph topology without gold labels, entity names, or dataset-specific relation mappings. They are not assertions of therapeutic mechanism. Shared equal assigns a unit coefficient to every signed feature. Shared learned uses $s(c,q)=\mathbf{w}^\top\mathbf{x}(c,q)$ and solves

$$\min_{\mathbf{w}\geq 0} \sum_{(q,c^+,c^-)\in\mathcal{P}} a_{q,+,-}\log\Big(1+\exp[-\mathbf{w}^\top(\mathbf{x}(c^+,q)-\mathbf{x}(c^-,q))]\Big)+\frac{\lambda}{2}\|\mathbf{w}\|_2^2.$$

The pair weights sum to one, balance recoverable positive rows and their eligible negatives, and give equal total weight to each source dataset in pooled transfer fitting. Known alternative positives for the same calibration-

training query are excluded as negative pairs. Absent-gold rows cannot create positive pairs, but remain in the validation-selection and test-metric denominators. We use 250 deterministic projected-gradient steps, five-fold cross-validation grouped by query ID, and $\lambda \in \{0.001, 0.01, 0.1, 1\}$. The highest mean validation-fold MRR is selected, preferring larger regularization in a tie. Equal candidate scores preserve original rank. An unconstrained fit tests the nonnegativity assumption; feature ablations refit the scorer and repeat regularization selection after removing each feature.

S12.2 Original-formula calibration and transfer protocols

To separate changes to coefficients from changes to features, a native-formula grid retains the dataset-specific support features in Sections S1–S2. It searches shortcut penalty $\{0, 0.1, 0.25, 0.5, 0.75, 1, 2\}$, contradiction penalty $\{0, 0.1, 0.2, 0.5\}$, and rank-prior mixture $\{0, 0.25, 0.5, 0.75, 0.8, 0.9, 0.95, 1\}$: 224 configurations per dataset. Selection maximizes validation MRR, breaking ties by larger rank-prior mixture and then smaller penalties. The fixed native control uses $(0.5, 0.1, 0)$ on every dataset. A score with shared features is a complementary baseline rather than merely a refit of the native feature family.

Table S16 Native-formula controls and unconstrained shared learning: test candidate MRR@20. The fixed native control shares coefficients but retains dataset-specific feature definitions. The unconstrained column uses the shared five-feature representation.

Dataset	Original	Fixed native	Grid selected	Unconstrained
PrimeKG	0.12533	0.12533	0.12533	0.12533
PharmKG	0.02816	0.02816	0.02816	0.03188
Hetionet	0.09544	0.09544	0.09544	0.10100
DRKG	0.04656	0.03982	0.04652	0.05811
repoDB	0.04811	0.02776	0.04814	0.06490

LODO fitting withholds the target dataset's entire validation label set. Regularization is selected by leave-one-source-dataset-out validation on the other four datasets, macro-averaging their validation MRRs, before refitting on the pooled sources. The target graph, structural candidate generator, and prepared subgraphs remain available. This assesses coefficient transfer, not zero-shot KGC. In the 20 directed transfer experiments, a scorer fitted on one dataset is evaluated on each of the other four datasets without target fitting. Validation and test can share query entities in the existing transductive benchmarks; grouped calibration cross-validation does not turn this into an unseen-disease evaluation.

Table S17 Selected shared-feature coefficients. Target uses the target validation pool; LODO excludes its labels. Features w_1–w_5 correspond to the vector above, including the explicitly negative direct-contact feature.

Dataset	Fit	λ	w_1	w_2	w_3	w_4	w_5
PrimeKG	Target	1.000	0.0621	0.0985	0.0000	0.0006	0.1481
PrimeKG	LODO	0.001	0.3724	0.0000	0.8965	0.0000	1.4384
PharmKG	Target	1.000	0.0121	0.0000	0.0120	0.0055	0.0512
PharmKG	LODO	0.100	0.2293	0.2176	0.0675	0.0200	0.6949
Hetionet	Target	0.010	0.0000	0.0000	0.0000	0.0000	2.0437
Hetionet	LODO	1.000	0.0495	0.0552	0.0234	0.0041	0.0914
DRKG	Target	0.001	1.8181	1.0539	0.0000	1.2828	1.9373
DRKG	LODO	0.001	0.4760	0.2054	0.0000	0.0000	1.9021
repoDB	Target	0.100	0.2297	0.0000	0.1239	0.0000	0.6921
repoDB	LODO	1.000	0.0334	0.0413	0.0139	0.0039	0.0881

S12.3 Candidate-ranking results and uncertainty

The main-text candidate table reports all five datasets and the equal-weight control. Shared learned matches the original formula on PrimeKG and raises its point estimate on the other four datasets. The correct feature-matched comparison is learned versus shared equal: the latter already raises candidate MRR over the original

formula on PharmKG, Hetionet, DRKG, and repoDB. The paired differences and their intervals are given in Table S18. LODO performs competitively but does not dominate equal weights, particularly on DRKG. These results support a reusable scoring representation while bounding the benefit attributable specifically to learned coefficients.

Table S18 Candidate MRR differences relative to shared equal weights. Intervals use 2,000 paired query-cluster bootstrap resamples, keeping multiple gold rows for one query together. They are exploratory 95% intervals without multiplicity correction.

Dataset	Fit	ΔMRR	95% lower	95% upper
PrimeKG	Learned	0.000138	-0.000132	0.000446
PrimeKG	LODO	0.000916	-0.000988	0.003898
PharmKG	Learned	-0.000007	-0.000067	0.000044
PharmKG	LODO	-0.000022	-0.000088	0.000031
Hetionet	Learned	0.014015	-0.007576	0.043253
Hetionet	LODO	0.012311	-0.009688	0.042107
DRKG	Learned	-0.000930	-0.003807	0.001440
DRKG	LODO	-0.002814	-0.008688	0.001746
repoDB	Learned	0.007536	0.000402	0.015176
repoDB	LODO	0.007536	0.000402	0.015176

Table S19 Directed transfer of learned candidate coefficients: test MRR@20. Rows identify the calibration source and columns the evaluation target. Diagonal entries are within-dataset fits; the 20 off-diagonal entries use no target calibration labels.

Source / target	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
PrimeKG	0.12533	0.03186	0.11463	0.05857	0.06480
PharmKG	0.12244	0.03188	0.10853	0.06200	0.06323
Hetionet	0.12079	0.03186	0.11528	0.05849	0.06490
DRKG	0.12537	0.03188	0.10467	0.06050	0.06424
repoDB	0.12146	0.03186	0.11542	0.05866	0.06490

S12.4 Calibration-label efficiency, ablation, and sensitivity

The label-efficiency experiment samples 10%, 25%, 50%, and 100% of validation query groups, with five seeded group orderings and nested regularization selection inside each subset. This yields 100 fits across all datasets. Table S20 reports test MRR means and sample standard deviations. PharmKG benefits from a larger calibration pool, whereas PrimeKG is unchanged across the tested fractions. Hetionet and repoDB are not monotonic. The standard deviations describe fitting variability, not uncertainty on independent test populations.

Table S20 Calibration-label efficiency: test MRR mean ± standard deviation over five subset/CV seeds. The column percentage is the fraction of available validation query groups, not the fraction of test data.

Dataset	10%	25%	50%	100%
PrimeKG	0.1253 ± 0.0000	0.1253 ± 0.0000	0.1253 ± 0.0000	0.1253 ± 0.0000
PharmKG	0.0290 ± 0.0043	0.0306 ± 0.0023	0.0317 ± 0.0004	0.0319 ± 0.0000
Hetionet	0.1131 ± 0.0047	0.1111 ± 0.0058	0.1090 ± 0.0058	0.1153 ± 0.0000
DRKG	0.0570 ± 0.0048	0.0582 ± 0.0004	0.0588 ± 0.0007	0.0605 ± 0.0000
repoDB	0.0656 ± 0.0035	0.0646 ± 0.0007	0.0642 ± 0.0009	0.0649 ± 0.0000

Table S21 Shared-feature ablation: test candidate MRR after refitting. "No shared" removes shared-neighbor count; "No direct" removes the signed direct-contact penalty. Each ablation repeats validation regularization selection.

Dataset	Full	No rank	No contact	No degree	No shared	No direct
PrimeKG	0.12533	0.12405	0.12614	0.12533	0.12533	0.06568
PharmKG	0.03188	0.03182	0.03188	0.03188	0.03186	0.02618
Hetionet	0.11528	0.11528	0.11528	0.11528	0.11528	0.09945
DRKG	0.06050	0.05973	0.06229	0.06050	0.05882	0.04339
repoDB	0.06490	0.07167	0.06490	0.06490	0.06490	0.04814

Coefficient sensitivity multiplies each fitted coefficient by an independent log-normal factor $\exp(\epsilon)$, with $\epsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma \in \{0.05, 0.1, 0.25, 0.5\}$. Five seeds per setting and two coefficient families yield 200 evaluations. The native rank-prior mixture is capped at one; zero coefficients remain zero. This is a coefficient-level perturbation, distinct from the per-candidate score noise in Section S8. Table S22 summarizes moderate and larger perturbations. Stability under these perturbations does not establish robustness to arbitrary feature corruption or missing relations.

Table S22 Coefficient perturbation: test MRR mean $\pm$ standard deviation over five seeds, and the fraction with unchanged top-ranked candidate relative to the unperturbed family.

Dataset	Family	MRR ($\sigma = 0.1$)	MRR ($\sigma = 0.5$)	Top1 same ($\sigma = 0.5$)
PrimeKG	Original	0.1253 ± 0.0000	0.1253 ± 0.0000	1.000
PrimeKG	Learned	0.1253 ± 0.0000	0.1253 ± 0.0000	1.000
PharmKG	Original	0.0282 ± 0.0000	0.0282 ± 0.0000	1.000
PharmKG	Learned	0.0319 ± 0.0000	0.0319 ± 0.0000	1.000
Hetionet	Original	0.0954 ± 0.0000	0.0948 ± 0.0015	0.982
Hetionet	Learned	0.1153 ± 0.0000	0.1153 ± 0.0000	1.000
DRKG	Original	0.0467 ± 0.0005	0.0430 ± 0.0017	0.701
DRKG	Learned	0.0602 ± 0.0001	0.0600 ± 0.0005	0.960
repoDB	Original	0.0453 ± 0.0035	0.0406 ± 0.0073	0.865
repoDB	Learned	0.0649 ± 0.0000	0.0646 ± 0.0007	0.949

S12.5 Frozen-model generation and automatic edge calibration

All candidate-order variants retain the full source graph and the same frozen dataset-specific checkpoint. The graph-selection arm instead retains the original fuzzy input order (raw/display-control order for repoDB) and its per-query edge budget. Exact generated-answer accuracy is reported separately from the promote-to-front adjusted MRR defined in the main text. An invalid output is treated as a virtual prediction before the list: a present gold is displaced by one rank and may still receive adjusted reciprocal-rank credit; an absent gold receives zero. Normalization takes the first line after any final "Answer: prefix, strips surrounding whitespace, quotes and punctuation, collapses whitespace, and ignores case. Exact matching accepts supplied display or canonical candidate names; ambiguous matches and substring-only matches are invalid. The deterministic decoder and matching rule are shared by Sections S12–S13 and the main module comparison.

Table S23 Candidate-order intervention with a full source graph. Panel A reports exact generated-answer accuracy (%); Panel B reports adjusted MRR@20. All 500/500/100/500/500 test rows are included.

A. Exact accuracy (%)

Candidate order	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
Raw	0.20	0.00	2.00	1.00	0.20
Original soft	0.00	0.00	3.00	1.40	0.20
Native grid	0.00	0.00	3.00	1.40	0.20
Shared equal	0.00	0.20	1.00	0.80	0.80
Shared learned	0.00	0.20	1.00	2.00	0.00
Shared LODO	0.00	0.20	2.00	1.80	0.00

B. Adjusted MRR@20

Candidate order	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
Raw	0.04767	0.01558	0.08192	0.04095	0.03664
Original soft	0.07468	0.02070	0.10009	0.04628	0.03662
Native grid	0.07461	0.02070	0.10009	0.04637	0.03664
Shared equal	0.07469	0.02288	0.07834	0.04677	0.04660
Shared learned	0.07461	0.02287	0.08578	0.05779	0.04400
Shared LODO	0.07478	0.02286	0.09194	0.05498	0.04400

Improved candidate MRR does not guarantee more exact generated answers. For repoDB, shared equal yields 4/500 exact answers and shared learned yields 0/500; the paired learned-minus-equal difference is -0.8 percentage points with exploratory 95% interval $[-1.702, -0.190]$. For DRKG, the corresponding accuracies are 0.8% and 2.0%, and the 1.2-point difference has interval $[0.000, 2.469]$. These low counts and mixed outcomes limit claims about generation gains even where candidate ranks improve.

The common edge scorer uses five features: candidate rank band, candidate incidence, query incidence, a negative direct query–candidate indicator, and bounded endpoint degree. It omits dataset-specific relation-family priorities. Twelve predefined coefficient vectors include fixed/equal controls and seeded perturbations of a common default. Each vector is evaluated on the full validation split; selection maximizes exact-answer accuracy, then adjusted MRR, then prefers the earliest vector. Source-only edge LODO selects by macro-validation performance over the other datasets. The source-only restriction applies to this new coefficient selection, while candidate order and graph budgets are inherited from the original pipeline. No test result is used to resolve a tie.

Table S24 Automatic edge calibration. A: selected zero-based trial IDs and number tied for best target validation accuracy and adjusted MRR. B–C: full-test results at fixed candidate order and per-query edge count.

A. Validation selection

Dataset	Auto trial	LODO trial	Best ties / 12
PrimeKG	1	2	6
PharmKG	3	2	2
Hetionet	0	2	12
DRKG	0	2	12
repoDB	2	3	1

B. Adjusted MRR@20

Dataset	Original fuzzy	Shared fixed	Shared auto	Edge LODO
PrimeKG	0.074687	0.074687	0.074676	0.074676
PharmKG	0.020971	0.020956	0.021317	0.020969
Hetionet	0.100087	0.100087	0.100087	0.100087
DRKG	0.046285	0.046285	0.046285	0.046285
repoDB	0.037776	0.037795	0.038159	0.036653

C. Exact accuracy (%)

Dataset	Original fuzzy	Shared fixed	Shared auto	Edge LODO
PrimeKG	0.00	0.00	0.00	0.00
PharmKG	0.20	0.20	0.20	0.20
Hetionet	3.00	3.00	3.00	3.00
DRKG	1.40	1.40	1.40	1.40
repoDB	0.40	0.40	0.40	0.20

All 12 vectors tie on both selection metrics for Hetionet and DRKG; this does not identify a unique optimum or prove equivalence on other graphs. The four compressed graph arms have the same exact accuracy on the first four datasets, while repoDB edge LODO gives 0.2% versus 0.4% for the other three. Thus automated selection makes the coefficient choice reproducible but does not establish a uniform downstream improvement. The complete configuration traces, learning curves, all perturbation levels, paired comparisons, and predictions are available through the accompanying repository's calibration scripts and result workbook.

S13 Information Retention and Alternative Subgraph Selection

S13.1 Evaluation design and selection policies

We assess evidence compression at three separate levels: retained graph structure, the representations supplied by the frozen graph encoder, and generated answers. All comparisons fix the candidate identities and their ordering to the original fuzzy input. For repoDB this is the original raw/display-control ordering. Candidate ranking metrics are therefore invariant by construction and are not used as evidence of information retention.

Source graphs and matched budgets. We distinguish the multiset of retrieved edge occurrences from the set of unique directed, relation-labeled triples. Two uncompressed references retain either the full multiset or only unique triples. The original fuzzy graph is read directly from its archived artifact. For each query, all compressed alternatives select exactly the number of unique triples in that artifact from the same unique source-edge set. The empty-graph diagnostic removes local message passing while retaining frozen global entity embeddings. These uncompressed and empty references are not treated as matched-budget competitors.

Alternative selection strategies. The alternatives include uniform random sampling without replacement; random ordering followed by candidate/query coverage; top-B native-score selection without coverage; native-score selection with a strict coverage-then-fill pass; endpoint-degree top-B; query-centered breadth-first ordering; query-personalized PageRank; terminal-personalized PageRank; shortest-path preservation; and relation-stratified selection. Native scores are recovered from archived features/scores, with exact triple and candidate alignment checks. The strict native-score coverage control is not claimed to reproduce every dataset-specific reservation pass of the original implementation.

PageRank uses an unweighted undirected source graph, damping 0.85, and either the query alone or the query and present candidates as uniformly weighted teleportation nodes. Dangling mass returns to the teleportation distribution. Iteration stops at an L1 change below 10^{-10} or after 200 iterations. Edges are ranked by the sum of endpoint scores. The shortest-path selector considers candidates in their fixed order, adds one shortest path of length at most three if the complete path fits the remaining budget, and fills remaining slots by native score. Relation-stratified selection cycles over relation IDs, ranking edges within each stratum by native score. Stable source-index tie-breaking is used unless the explicitly seeded random ordering applies. No selector accesses the held-out gold label.

Five random seeds are evaluated at the native budget. Additional strict-budget curves use 20%, 35%, 55%, 75%, and 90% of unique source edges, with three seeds for random selection. The main CPU study contains 62 variants per dataset/split across all five datasets and both validation and test. A separate 60-row native-implementation audit calls the original selector functions, reproduces the archived edge sets at their native budgets, and varies only the requested budget. It preserves and reports any budget excess caused by the original reservation passes rather than silently imposing a new policy.

Structural retention definitions. Candidate coverage measures incidence to at least one retained edge and is reported both absolutely and relative to source-covered candidates, including top-, middle-, and tail-rank bands. Query coverage, unique-node retention, relation-type retention, and direct query–candidate exposure are reported separately. For $k \in \{2, 3\}$, let $C_{\le k}(G_q)$ be candidates reachable from the query within k edges in the source graph and let $P_k(G_q)$ be its edge-labeled simple path witnesses of exactly k edges ending at a candidate. We compute

$$R_{\mathrm{reach},k}(q) = \frac{|C_{\le k}(S_q)|}{|C_{\le k}(G_q)|}, \qquad R_{\mathrm{path},k}(q) = \frac{|\{p \in P_k(G_q) : E(p) \subseteq E(S_q)\}|}{|P_k(G_q)|}.$$

The walks are traversable in either direction; exact edge identities, relation labels, and traversal signs are retained. Consequently these quantities measure structural connectivity, not necessarily directed biochemical pathways. Typed pattern retention measures whether each source relation/sign sequence of length two or three still has at least one witness. A separate audit measures retention of unique incoming edges within two predecessor steps of the query and candidates, matching the two message-passing layers of the actual directed encoder more closely.

Relation-frequency and node-incidence mass distributions are compared using base-2 Jensen–Shannon divergence. Rare-relation edge recall uses relations with source frequency at or below the per-query median.
Any retention with a zero source denominator is undefined, omitted from the mean, and accompanied by its eligible sample count. The reported means are averages of per-query-row ratios, not ratios of pooled path counts. Quantiles, complete-loss fractions, source-query-presence strata, and gold-coverage strata supplement the aggregate results.

Frozen-model representation and answer evaluation. The GPU study evaluates all 22 native-budget variants and six additional strict-budget variants (native-score coverage, terminal PageRank, and random seed zero at 35% and 75%) on the complete test split. The same frozen checkpoint, prompt, candidate order, and deterministic decoding settings apply to every selector within each dataset. We measure cosine similarity and relative L2 change of query/candidate adapter outputs against both source references. Generated-answer exact accuracy, invalid-output rate, top-candidate copying, normalized output agreement, and the explicitly standardized promote-to-front adjusted MRR are reported separately.
The existing encoder falls back to global/global embeddings for graphs with at most ten edge occurrences; we explicitly report this condition at small budgets.

Uncertainty and scope. Paired query-cluster bootstrap intervals use 2,000 resamples so that rows sharing the same query remain together. Random-selector means additionally receive joint query-cluster/seed bootstrap intervals; seed standard deviations are reported separately. These are exploratory intervals without multiplicity adjustment. All methods, ratios, and seeds are declared before inference; no test-winning configuration is promoted to a new default. Retention is measured relative to retrieved source subgraphs and does not establish preservation of all KG facts, expert-validated mechanisms, or clinical utility.

S13.2 Structural retention results and compression trade-offs

The primary CPU study completed 620 dataset/split/variant summaries, with a further 60 summaries from the native-implementation budget audit. All native edge sets were reproduced exactly at their archived budgets on validation and test. Alternative selectors preserved candidate order and satisfied the matched unique-edge budgets. These summaries reuse 2,100 validation and 2,100 test rows; they are not independent new datasets or examples.

Duplicate removal is distinct from unique-edge removal. PrimeKG source graphs contain a mean of 59.932 edge occurrences but only 38.094 unique triples per test row. The original fuzzy output retains 32.340 unique triples. Thus, the mean reduction of 27.592 occurrences consists of 21.838 duplicate occurrences and 5.754 unique triples. Approximately 79.1% of the aggregate count reduction is attributable to duplicate removal. The other four datasets have no such source duplicates in these prepared test inputs. This is a counting decomposition, not a claim that duplicates have no computational effect: the encoder can process repeated triples as repeated messages.

Table S25 Full matched-budget comparison of three-hop reachability retention. Random rows average all five seeds. Eligible row counts are 488, 500, 90, 386, and 297 for PrimeKG, PharmKG, Hetionet, DRKG, and repoDB. Candidate order and unique-edge count are fixed per query.

Selector	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
Original fuzzy	0.7737	0.9010	0.4835	0.8921	0.9546
Native score top-k	0.7805	0.5600	0.1455	0.0922	0.0000
Strict native coverage	0.7737	0.9069	0.4835	0.6242	0.5634
Degree top-k	0.8408	0.5532	0.6251	0.5734	0.2591
Query BFS	0.8328	0.6254	0.7047	0.8075	0.9164
Query PPR	0.7868	0.6021	0.6814	0.7940	0.9005
Terminal PPR	0.9111	0.5972	0.6251	0.7305	0.9112
Shortest paths	0.9985	1.0000	1.0000	1.0000	1.0000
Relation stratified	0.7694	0.8429	0.8479	0.6974	0.9399
Random (5 seeds)	0.7760	0.4423	0.4848	0.5043	0.5465
Random + coverage (5 seeds)	0.7816	0.7465	0.6068	0.7125	0.7700

Coverage does not imply path preservation. Although original fuzzy retrieval preserves relative candidate coverage on all five datasets, its reachability retention varies substantially (Table S25). Fuzzy retrieval preserves substantially more reachability than uniform random selection on PharmKG, DRKG, and repoDB, whereas the PrimeKG and Hetionet point estimates are similar to random. The joint query-cluster/seed bootstrap interval for random minus fuzzy reachability is $[-0.4829, -0.4337]$ on PharmKG. The corresponding intervals are $[-0.4433, -0.3353]$ on DRKG and $[-0.4492, -0.3661]$ on repoDB; the PrimeKG and Hetionet intervals include zero. These are exploratory, unadjusted intervals.

The shortest-path baseline retains nearly all source-reachable candidates at the same edge count. This is partly aligned with its explicit selection objective and does not establish semantic superiority. It also trades away coverage of other candidates: relative candidate coverage is 0.8514 on Hetionet, 0.8199 on DRKG, and 0.8971 on repoDB, compared with 1.0000 for original fuzzy retrieval. Thus, preserving candidate incidence and preserving connected query–candidate paths are different selection goals.

Table S26 Exact edge-labeled path retention at length three. N counts rows with at least one source path; zero-denominator rows are excluded. Random is mean $\pm$ seed standard deviation. Paths are structural witnesses, not validated therapeutic mechanisms.

Dataset	N	Fuzzy	Random (5 seeds)	Shortest paths
PrimeKG	488	0.6543	0.6443 ± 0.0047	0.8199
PharmKG	31	0.0000	0.1801 ± 0.0967	0.8604
Hetionet	64	0.0000	0.1615 ± 0.0589	0.6207
DRKG	201	0.4767	0.1629 ± 0.0078	0.6496
repoDB	219	0.5286	0.1552 ± 0.0185	0.8322

Length-three path retention exposes additional losses that are hidden by coverage alone (Table S26). Original fuzzy retrieval retains no such witnesses in the eligible PharmKG and Hetionet rows. This does not mean that every row loses all connectivity: only 31 PharmKG rows have a length-three source witness, and shorter paths can remain. Conversely, original fuzzy retrieval retains more length-three witnesses than random on DRKG and repoDB. Table S27 reports complementary structural diagnostics. Rank-band and conditional diagnostics remain available in the accompanying workbook.

Table S27 Complementary retention diagnostics for original fuzzy retrieval. Nodes and relation types are retained fractions; relation JSD is base-2 Jensen–Shannon divergence (zero means identical distributions). Incoming-2 measures directed incoming context within two predecessor steps. Rare edges belong to relations with frequency at or below the source median.

Dataset	Nodes	Relation types	Relation JSD	Incoming-2	Rare edges
PrimeKG	0.9402	0.9378	0.0046	0.8593	0.8069
PharmKG	0.5735	0.8984	0.1375	0.6180	0.8774
Hetionet	0.5868	0.8370	0.0438	0.4309	0.5330
DRKG	0.5701	0.6562	0.0708	0.6100	0.4644
repoDB	0.5102	0.8582	0.0319	0.5433	0.6319

Source availability. Fuzzy outputs contain the query node in 100%, 96.8%, 100%, 84.8%, and 59.4% of test rows on PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. Conditional reachability retention must be interpreted with its source eligibility rather than as a fraction of every test row.

Budget feasibility and operating curves. The native replay audit confirms that the original PrimeKG coverage-first implementation can exceed a newly imposed small target. At a requested budget of 20% of unique source edges, its mean requested/actual counts are 7.576/15.662, and 99.8% of test rows exceed the request. At 35%, the counts are 13.336/15.886 and the excess rate is 74.2%. No excess occurs at the archived native budgets used in the matched comparison. We report these native-policy points separately from strict-budget controls. Operating curves must therefore identify the selector and distinguish requested budget from actual graph size.

S13.3 Frozen-model representation and generated-answer results

The completed GPU study comprises 28 variants on each of five datasets, yielding 140 summaries and 58,800 row–variant evaluations of the same 2,100 test rows. Every comparison uses the same dataset-specific checkpoint and fixed candidate order. All methods use the same inference and exact-answer matching rules described in Section S12.

Table S28 Exact generated-answer accuracy (%) under the standardized matching rule. Random rows average all five seeds. Source raw/unique and empty local graph are diagnostic references with different budgets; the other rows use matched unique-edge budgets. The source multiset uses the original fuzzy input candidate order, so it is an order-matched graph reference rather than the raw-candidate baseline.

Selector	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
Source multiset	0.00	0.00	3.00	1.40	0.20
Source unique	0.00	0.00	3.00	1.40	0.20
Empty local graph	0.40	0.00	1.00	1.20	1.80
Original fuzzy	0.00	0.20	3.00	1.40	0.40
Native score top-k	0.00	0.00	2.00	1.20	2.20
Strict native coverage	0.00	0.20	3.00	1.40	0.40
Degree top-k	0.00	0.20	0.00	1.20	1.60
Query BFS	0.00	0.20	0.00	1.20	0.40
Query PPR	0.00	0.20	0.00	1.20	0.40
Terminal PPR	0.00	0.20	2.00	1.20	0.40
Shortest paths	0.00	0.20	3.00	1.40	0.20
Relation stratified	0.00	0.00	2.00	1.20	0.20
Random (5 seeds)	0.04	0.04	3.00	1.24	0.52
Random + coverage (5 seeds)	0.00	0.08	3.00	1.40	0.16

Table S29 Original fuzzy versus the uncompressed source: encoder output changes and normalized generated-answer agreement. Candidate metrics average the 20 candidate embeddings within each row, then average rows. A cosine near one does not exclude a change in vector magnitude; relative L2 is reported separately.

Dataset	Query cosine	Candidate cosine	Query rel. L2	Candidate rel. L2	Agreement (%)
PrimeKG	0.8828	0.9453	0.8818	0.1013	78.8
PharmKG	0.8718	0.9991	0.6130	0.0032	80.0
Hetionet	0.9209	1.0000	0.3259	0.0000	100.0
DRKG	0.9312	0.9867	0.3834	0.0526	100.0
repoDB	0.9814	0.9806	0.1326	0.0862	80.8

Structural loss and output invariance are distinct. On Hetionet, original fuzzy retrieval preserves the candidate encoder outputs (mean cosine 1.0000, relative L2 0.0000) and all generated answers relative to the source graph. Nevertheless, query outputs change (mean cosine 0.9209, relative L2 0.3259), and all eligible length-three path witnesses are lost. Thus even exact candidate-output agreement cannot be interpreted as whole-graph information preservation. Directed message passing, global embeddings, and which nodes are injected into the prompt all affect how a structural change propagates to generation.
DRKG likewise has 100% generated-answer agreement for fuzzy versus source, although mean query and candidate cosines are 0.9312 and 0.9867; their relative L2 changes are 0.3834 and 0.0526. On PrimeKG and PharmKG, answer agreement is only 78.8% and 80.0%, respectively. Their candidate cosines are 0.9453 and 0.9991, while query cosines are lower at 0.8828 and 0.8718. These results show why candidate similarity alone is insufficient to characterize information supplied to the LLM.
On repoDB, fuzzy/source answer agreement is 80.8%, with query/candidate cosines of 0.9814/0.9806. Fuzzy generates 2/500 exact answers, compared with 1/500 for source raw, 11/500 for native-score top-k, and 9/500 for the empty-local-graph reference. The latter retains frozen global embeddings; it is not a knowledge-free model. These counterexamples are reported without selecting a new method using test outcomes.

Downstream differences do not establish a universal winner. The exact-accuracy comparison does not show consistent superiority of fuzzy over all alternative selectors. Several differences correspond to only one correct answer in a 500-row test set. For example, fuzzy produces 0/500, 1/500, 3/100, and 7/500 correct answers on PrimeKG, PharmKG, Hetionet, and DRKG, respectively. The joint query/seed and paired query bootstrap tables are supplied alongside these point estimates; lack of a detected difference is not a demonstration of equivalence. These low counts limit conclusions about small downstream improvements.

S13.4 Budget–retention curves

Figures S4–S6 summarize edge-count decomposition and strict-budget operating curves. The requested ratios apply to unique source triples. The original dataset-specific policy and the strict coverage control are distinguished throughout; neither preserved candidate ranking nor an unchanged generated answer certifies preservation of the source evidence.

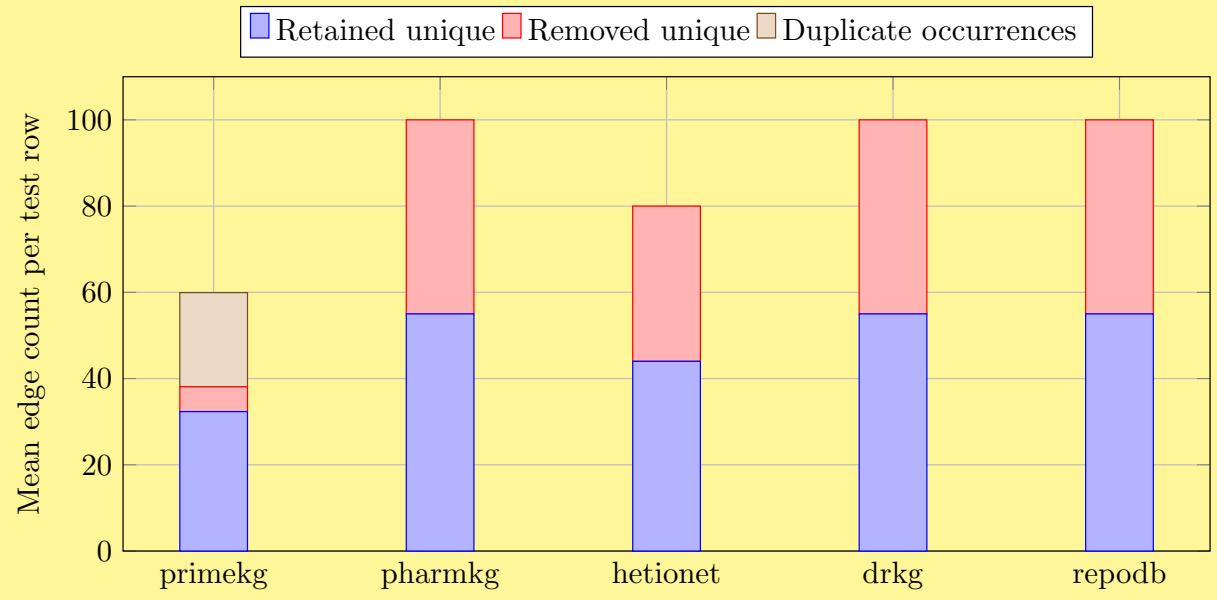


Fig. S4 Decomposition of source edge occurrences into retained unique triples, removed unique triples, and duplicate occurrences. Counts are means over test rows; duplicate removal is separated from structural edge loss.

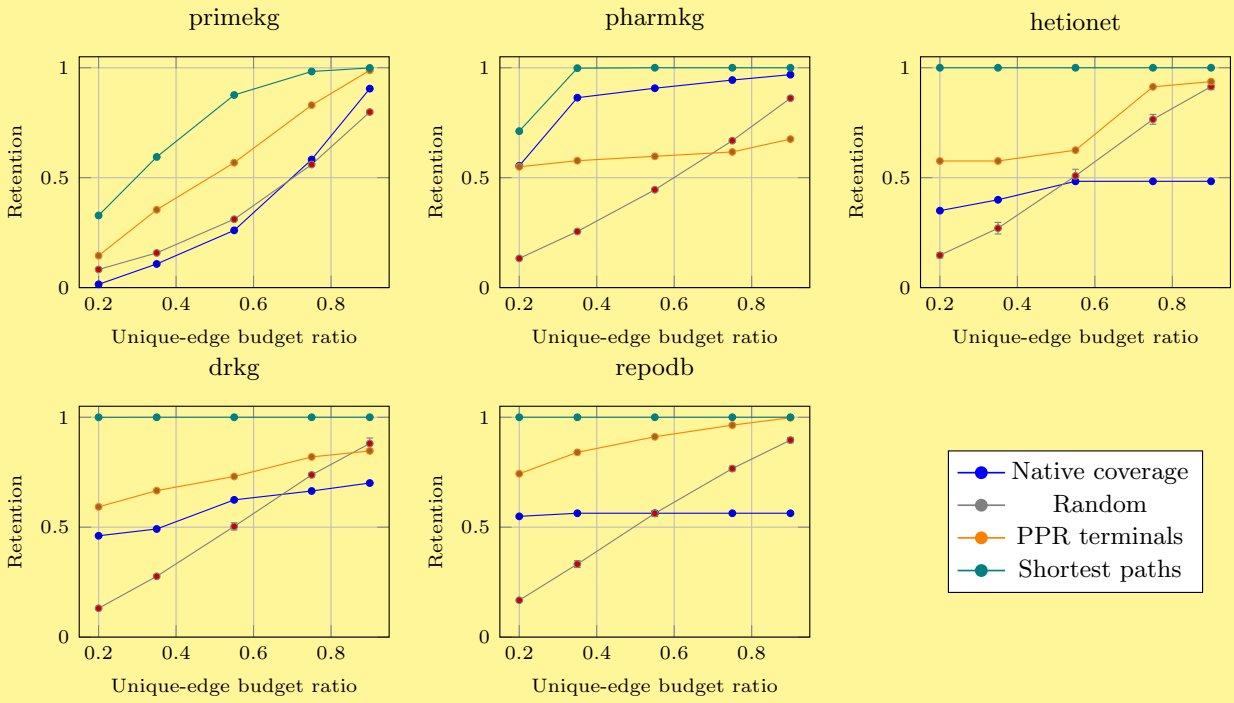


Fig. S5 Candidate–query reachability retention for strict-budget selectors. The native-score coverage control is distinguished from the original dataset-specific selector. Random error bars show seed variability, not test confidence intervals. Native implementation budget violations are reported separately in the audit tables.

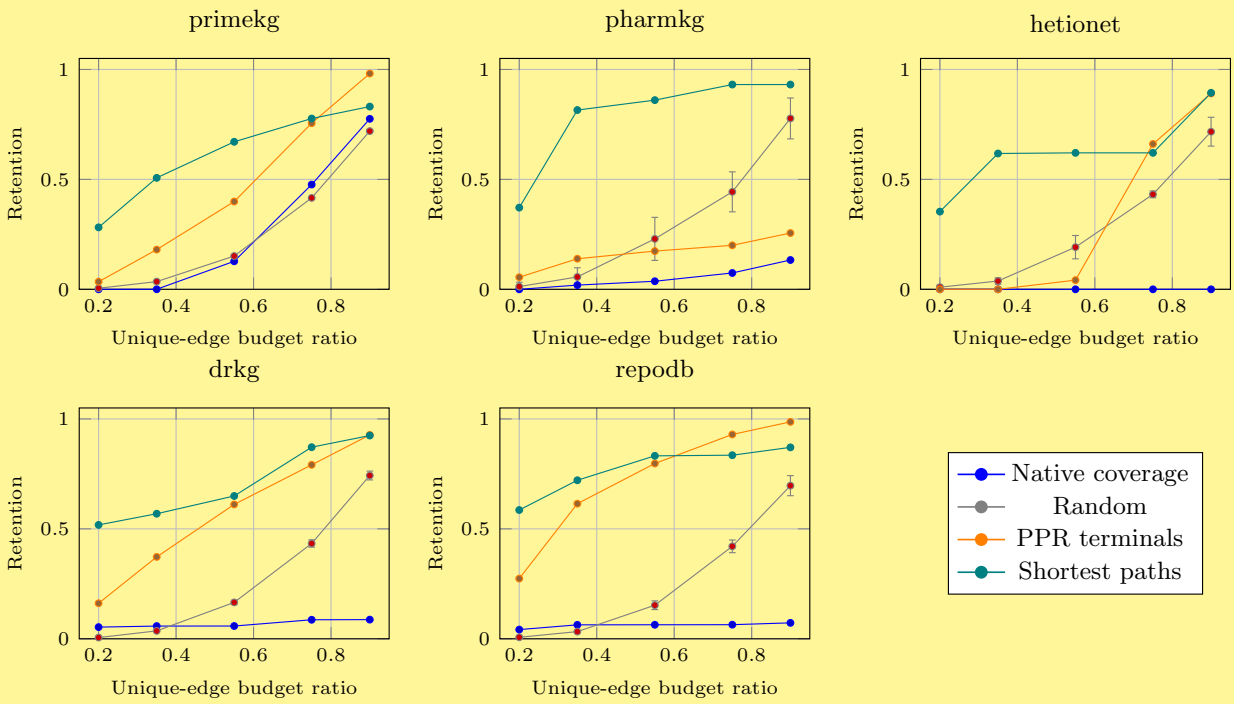


Fig. S6 Exact length-three path-witness retention under the same strict-budget controls. Means exclude queries without a corresponding source witness; eligible counts are provided in the accompanying tables.